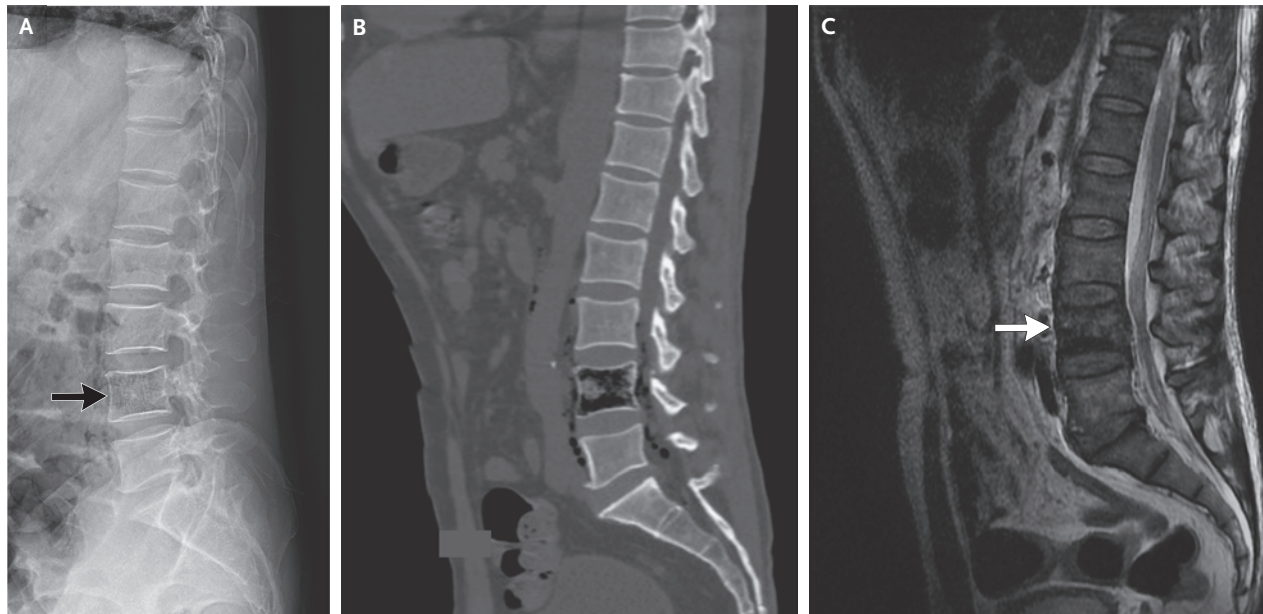


IMAGES IN CLINICAL MEDICINE

Stephanie V. Sherman, M.D., *Editor*

Emphysematous Vertebral Osteomyelitis



A 62-YEAR-OLD MAN PRESENTED TO THE EMERGENCY DEPARTMENT WITH a 5-day history of low back pain and a 1-day history of lethargy. Physical examination was notable for tenderness over the lumbar spine and a normal neurologic examination. Radiographs of the lumbar spine showed gas within L4 (Panel A, arrow). Computed tomography of the lumbar spine confirmed the presence of intramedullary gas within L4 — a finding known as the pumice stone sign — and also showed that the gas extended into the surrounding soft tissue, epidural space, and psoas muscles (Panel B). Magnetic resonance imaging of the lumbar spine showed the intramedullary gas, as well as a gas-containing epidural abscess that was causing mild compression of the cauda equina (Panel C, arrow). Cultures of blood, sputum, and urine grew extended-spectrum β -lactamase-producing *Klebsiella pneumoniae*. A diagnosis of emphysematous vertebral osteomyelitis and epidural abscess due to disseminated klebsiella infection was made. Although surgery is the standard treatment for this type of infection, a multidisciplinary team recommended medical management for this patient owing to his critical illness, the stability of his spine, and the absence of neurologic deficits. The patient received an extended course of antibacterial agents. At a 3-month follow-up visit, he was doing well.

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